

Advanced Machine Learning

DATA 642

Lecture 10 — Clustering for Image Segmentation

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Central Idea of Clustering for Image Segmentation

Partition an image into distinct regions or segments based on similarity in pixel values.

- **Methodology:**

- Clustering algorithms group pixels together based on color or intensity similarity.

- **Purpose:**

- Extract meaningful structures or objects from the image.
- Enable tasks such as object detection, tracking, and analysis.



- **Computer Vision:** Clustering for image segmentation is extensively used in computer vision applications such as object recognition, scene understanding, and image classification. By segmenting images into meaningful regions, computer vision systems can analyze and interpret visual information more effectively.
- **Medical Imaging:** In medical imaging, clustering-based image segmentation techniques are employed for tasks such as tumor detection, organ segmentation, and medical image analysis. By segmenting medical images into distinct regions, clinicians can identify abnormalities, diagnose diseases, and plan treatment strategies more accurately.
- **Satellite Image Analysis:** Clustering algorithms are widely applied in satellite image analysis for land cover classification, urban growth monitoring, and environmental monitoring. By segmenting satellite images into different land cover types or regions, researchers can analyze changes in land use patterns, monitor environmental changes, and support urban planning efforts.
- **Various Other Domains Requiring Image Processing and Analysis:** Clustering-based image segmentation finds applications in various domains requiring image processing and analysis, including remote sensing, agriculture, robotics, autonomous vehicles, and quality control in manufacturing. It enables tasks such as object detection, feature extraction, and anomaly detection, contributing to advancements in diverse fields.

Using K -Means for Image Segmentation

- **Methodology:**

- Treat each pixel as a data point in a high-dimensional space (e.g., RGB color space).
- Apply K -Means clustering to group pixels into K clusters based on their color similarity.
- Assign each pixel to the cluster with the nearest centroid.

- **Result:**

- Each pixel is assigned to a cluster, effectively segmenting the image into regions of similar colors.
- Boundaries between segments represent areas of color transition.

- **Benefits:**

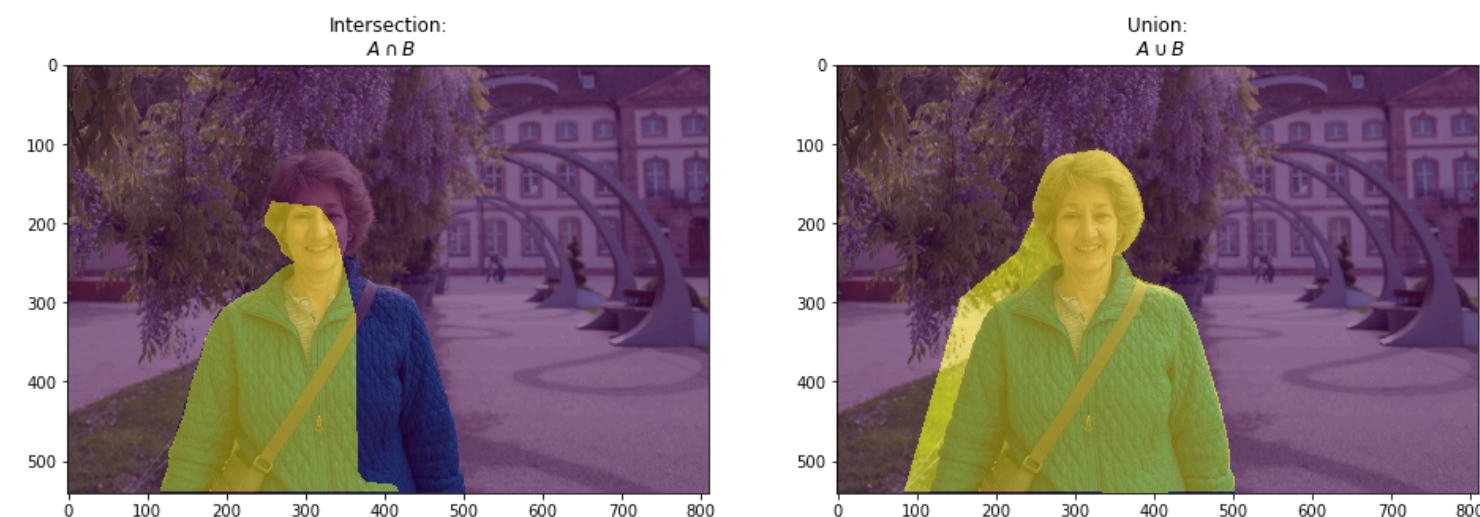
- Simplifies image analysis by separating regions of interest.
- Enables tasks such as object detection, image compression, and feature extraction.
- Offers a computationally efficient approach to image segmentation.

Evaluating Image Segmentation

- **Pixel Accuracy:** Measures the percentage of correctly classified pixels compared to ground truth.



- **Intersection over Union (IoU):** Computes the overlap between predicted and ground truth regions.



- **Dice Coefficient:** Quantifies the spatial agreement between segmented and ground truth regions.
- **Boundary Displacement Error (BDE):** Calculates the average distance between segment boundaries and ground truth edges.
- **Region Similarity:** Compares the similarity of segmented regions to ground truth using metrics such as Structural Similarity Index (SSI) or Mutual Information (MI).